

Sai Pavan Kumar Vedulapurapu

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Professional Summary

Data Science and Big Data Analytics student with strong foundations in Python, Machine Learning, Deep Learning, Statistics, and Data Analytics. Experienced in building AI-powered applications, REST APIs, analytics dashboards, and data-driven systems using FastAPI, Pandas, NumPy, and SQL. Passionate about solving real-world problems using Machine Learning and Generative AI technologies with strong analytical, communication, and collaborative skills.

Education

SRM University, AP

Aug 2023 – Present

B.Sc. (Hons.) in Computer Science — Big Data Analytics

CGPA: 8.64 / 10

Relevant Coursework: Machine Learning, Data Mining and Data Warehousing, Database Management Systems, Data Structures, Analysis of Algorithms, Operating Systems, Statistics for Data Science

Technical Skills

Programming Languages	Python, SQL, JavaScript
Data Science & AI	Machine Learning, Deep Learning, NLP, Generative AI, Statistics, Data Analytics
Libraries & Frameworks	Pandas, NumPy, Scikit-learn, FastAPI, Streamlit
Visualization & BI	Power BI, Excel, Data Visualization
Databases	PostgreSQL, MySQL, MongoDB, SQLite
Web & APIs	REST APIs, Next.js, React, Node.js
Tools & Platforms	Git, GitHub, Docker, Linux, Bash
Core CS Fundamentals	DBMS, Data Structures, Algorithms, Operating Systems

Certifications

MongoDB Certified Associate Developer

[View Certificate](#)

- Demonstrated proficiency in MongoDB data modeling, CRUD operations, indexing, and aggregation framework.
- Gained practical knowledge in NoSQL databases and backend application development.

Projects

GritOS: Intelligent Operational & Financial Analytics Platform

[GitHub Repository](#)

Technologies: Python, FastAPI, Next.js, Pandas, Generative AI

- Developed an AI-powered analytics platform to analyze operational and financial business data.
- Built REST APIs using FastAPI for data processing, forecasting, and analytics workflows.
- Implemented data-driven algorithms to simulate market scenarios and predict financial impacts.
- Integrated LLaMA 3 via Groq API to provide Generative AI-based business insights and recommendations.
- Collaborated on full-stack development, testing, and documentation of application components.

SentientQ: Quantum-Powered Sentiment Analysis Engine

[GitHub Repository](#)

Technologies: Python, NLP, FastAPI, PennyLane, Machine Learning

- Built a sentiment analysis application using NLP and Machine Learning techniques.
- Designed a Variational Quantum Circuit (VQC) model for text classification experiments.
- Developed APIs and dashboards for real-time sentiment monitoring and analytics visualization.
- Applied data preprocessing, feature extraction, and classification techniques on review datasets.

- Conducted testing and performance evaluation for model optimization and stability.

Fraud Detection and Analytics Platform

 [GitHub Repository](#)

Technologies: Python, Machine Learning, FastAPI, Streamlit, SQLite

- Developed a fraud detection platform for transaction analysis and anomaly identification.
- Built scalable FastAPI backend services for transaction scoring and predictive analytics.
- Designed interactive dashboards using Streamlit for visualizing fraud patterns and reports.
- Worked with structured datasets, data simulation, and automated testing workflows.
- Improved analytical decision-making using real-time monitoring and classification techniques.

Achievements & Activities

Coding Club Member, SRM University AP

Aug 2024 – Present

- Participated in collaborative coding sessions, technical discussions, and problem-solving activities.
- Worked on peer learning initiatives focused on AI, Machine Learning, and software development.

Hackathon Participant — SWEeyam, QC², Code4Her, Schrödinger's Cat

- Built innovative prototypes and AI-based solutions in competitive hackathon environments.
- Collaborated with teams to design, develop, and present technical solutions under deadlines.